

Status	Finished
Started	Monday, 3 November 2025, 1:51 PM
Completed	Monday, 3 November 2025, 2:24 PM
Duration	33 mins 16 secs

Question **1**

Correct

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and $153 = 1^3 + 5^3 + 3^3$.

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and $123 \neq 1^3 + 2^3 + 3^3 = 36$.

Example 3:

Input:

1634

Output:

true

Note:

$1 \leq N \leq 10^8$

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<math.h>
3 int main() {
4     long long int num,sum=0,nod=0,rem,temp;
5     scanf("%lld",&num);
6     temp=num;
7     while(num>0)
8     {
9         nod++;
10        num=num/10;
11    }
12    num=temp;
13    while(num>0) {
14        rem=num%10;
15        sum=sum+pow(rem,nod);
16        num=num/10;
17    }
18    if(sum==temp)
19        printf("true");
20    else
21        printf("false");
22    return 0;
23 }
```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question **2**

Correct

Take a number, reverse it and add it to the original number until the obtained number is a palindrome.

Constraints $1 \leq \text{num} \leq 999999999$ **Sample Input 1**

32

Sample Output 1

55

For example:

Input	Result
32	55
1234	5555

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main() {
3      long long int num,sum,revnum,tempnum,tempsum;
4      scanf("%lld",&num);
5      while(1) {
6          revnum=0;
7          tempnum=num;
8          while(num) {
9              revnum=revnum*10+(num%10);
10             num=num/10;
11         }
12         sum=tempnum+revnum;
13         tempsum=sum;
14         revnum=0;
15         while(sum){
16             revnum=revnum*10+(sum%10);
17             sum=sum/10;
18         }
19         if(tempsum==revnum)
20
21             break;
22         num=tempsum;
23     }
24     printf("%lld",tempsum);
```

```
25 |     return 0;  
26 | }
```



	Input	Expected	Got	
✓	32	55	55	✓
✓	1234	5555	5555	✓

Passed all tests! ✓

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Question **3**

Correct

Maya, a student in an arts and crafts class, wants to create a pattern using stars (*) in a specific format. She plans to use a program to help her construct the pattern.

Write a program that takes an integer as input and constructs the following pattern using nested for loops.

Input: 5

Output:

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main() {
3      int n,i,j;
4      scanf("%d",&n);
5      for(i=1;i<=n;i++) {
6          for(j=1;j<=i;j++) {
7              printf("* ");
8          }
9          printf("\n");
10     }
11     for (i=n-1;i>=1;i--) {
12         for(j=1;j<=i;j++) {
13             printf("* ");
14         }
15         printf("\n");
16     }
17     return 0;
18 }
```



	Input	Expected	Got	
✓	5	<pre>* *</pre>	<pre>* *</pre>	✓

Passed all tests! ✓